5. Costs of Industrial Hygiene Control

Introduction

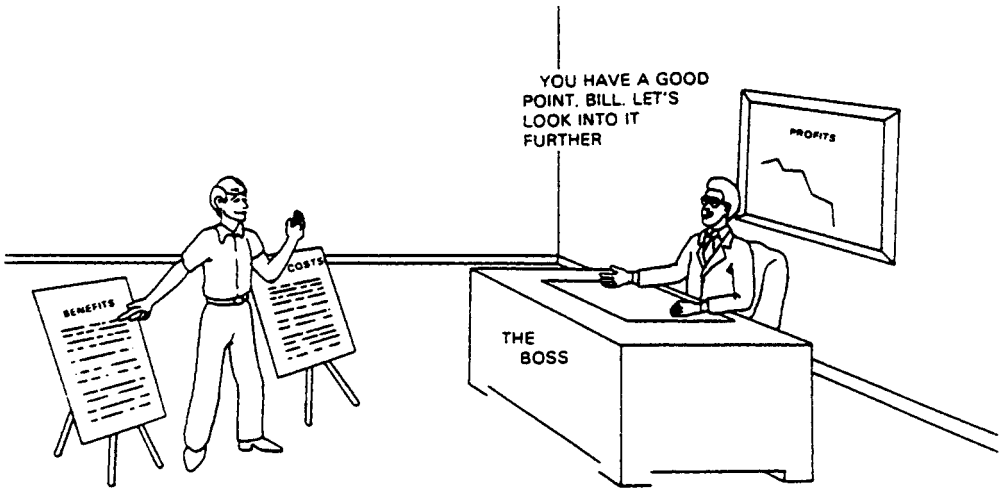
Because of an involvement and commitment to protecting the worker's health and safety, the industrial hygiene engineer often takes for granted that particular controls are needed to protect the worker. The fact that a worker may be exposed to a particular hazard is sufficient reason to justify the need for control. However, management may not have the same viewpoint. The objective of management is to maximize profits. Industrial hygiene controls cost money and do not necessarily result in added profit to the firm. Management is more likely to be desirous of considering the risks of not controlling a particular hazard versus the costs and effects on profit to obtain these controls.

One of the jobs of the industrial hygiene engineer is to convince management of the need for control. The industrial hygiene engineer must become a salesman of a product, the control of exposure to occupational health hazards. In order to do this, a logical decision process and approach to problems must be used in the same manner as management approaches other decisions. Since the economics of decisions are important to management in other areas, they must also be important to the when related to the implementation of a given control method.

In order to sell industrial hygiene controls to management, the industrial hygiene engineer must approach management on its own ground. A case must be made for why a particular control is required. The benefits must be clearly outlined as well as the costs that are involved. It may not always be possible to sell industrial hygiene control as a method for improving the firm's profitability. However, it may be possible to minimize the costs of the controls required and to outline those costs associated with the risks that are taken by not instituting the controls even though, on a purely economic basis, they do not pay for themselves. In other cases, the decision can be made on pure economics. Thus, it is important to have an understanding of the costs of industrial hygiene control and be able to utilize basic economic analysis tools to present the case.

A number of factors are important when selling an industrial hygiene control program to management. From the economic standpoint, the costs of control versus the costs of no control must be considered and compared. Other factors that enter the picture are the legal requirements as stipulated by the Occupational Safety and Health Act as well as any state or local provisions. Also, the moral responsibility of the employer to the worker must be considered. The industrial hygiene engineer must consider the risk factor of

Figure 8.5.1



exposure in order to determine the costs involved. Finally, the industrial hygiene engineer must be aware of the level of danger to the worker that results from a particular exposure. Is the level of danger extreme, resulting in death, disability, or physical impairment; is the danger moderate, resulting in illness with lost time; or is the danger low, resulting only in discomfort to the worker but not lost time or accompanying harm to the worker's health?

Economic Cost of Not Providing Controls

First, consider the costs of not providing controls for hazards. That is, what costs accrue to the organization when appropriate controls are not present, and an exposure occurs?

Workers' Compensation. The purpose of workers' compensation is to remove the risk of the loss of employment, as a result of injury, from the employee. Workers' compensation insurance provides a fund base from which to compensate a worker when a job-related accident or illness occurs. Prior to the institution of workers' compensation, such protection was provided only as a result of a common law suit filed by the employee with the employer as the defendant. This dependence upon common law provided little protection to the worker. Workers' compensation insurance was developed and exists throughout this country to provide such protection to the worker.

Workers' compensation laws are governed by the individual state in which the business resides. In general, most industries are required to have coverage of their employees. There is some variance among states as to the

number of employees and the type of industry that require coverage. The type of coverage and the payment schedule also vary among states. In most states, the law covers both occupational illness and injury.

The premium costs that must be paid by a particular industry for workers' compensation depend upon the risks of the industry group in which the company operates, the experience of the company in terms of illness and injury, and an adjusted annual gross payroll based upon a maximum amount of compensation per week for each employee. The basic rate is established for a particular industry group as a result of the experience of that industry group. The experience of a particular company is evaluated and compared to the industry group, and a rating is determined. This rating is called the experience modification factor. If the experienced loss has been less than the average for the industry over a historical period (generally three years), then the experience modification factor is less than 1. If the experienced loss has been the same as the average for the industry over the historical period, the experience modification factor equals 1. If the experienced loss is greater than the industry over the historical period, the experience modification factor is greater than 1. The premium cost is determined by multiplying the base rate times the annual payroll (calculated for determination of rate) and the experience modification factor.

For example, consider a firm in which the base industry rate is $1.65 per $100 payroll and which has a $360,000 adjusted annual payroll after deductions for wages in excess of the maximum annual rate. If the experience modification factor for this firm is 1.8, then the premium would be determined as follows:

Premium = $1.65/100 x 360,000 x 1.8
Premium = $10,692

The premium that results from a high experience modification factor is a cost of not providing controls. In terms of actual dollars, it is the difference between the industry average with an experience modification factor of 1 and the rate being paid. In the above example, the cost to the particular plant involved is $4,752. An additional cost involving workers' compensation is the cost of processing claims against the workers' compensation benefits.

Disability Insurance. Many companies pay for disability insurance in addition to premiums for workers' compensation. This fringe benefit continues to become more prevalent in industry. The cost of higher premiums for such insurance may be a cost of not providing industrial hygiene controls. In addition, any sick leave that is incurred but not insured and that results from occupational illness or injury is also a cost of not providing controls.

Legal Costs. Specific laws may be in existence to specify the requirement for controls. The absence of proper control may result in potential losses, either from law suits or from penalties assessed by enforcement agencies. The absence of controls can result in such costs being incurred by the organization.

Replacing Lost Employees. As a result of occupational illness or injury, employees may be lost to the organization either for short periods of time or, in the case of disability or death, forever. Assuming that such employees are productive and have been properly trained, the cost of replacing these employees is significant. Included in this cost factor are the costs for recruiting, interviewing, and hiring new employees and the cost of training these new employees to become productive. A hidden cost is the loss of production which results from the replacement of an older, well-trained employee with a new employee. In the case of short-term absence, the cost of overstaffing which is needed to provide for such short-term absence is also a part of this factor.

Medical Insurance. Medical insurance coverage is almost universally provided by employers today. In many cases, medical insurance premiums are a result of group experience. Increased medical costs as a result of occupational injury or illness can raise the cost of such premiums.

Availability of Labor Pool. The risk of illness and injury which is assumed by a worker on a particular job can affect the size of the labor pool available. If the job appears to be a high-risk job, few applicants will be willing to consider such a job. As the labor pool becomes smaller, it is necessary to raise the pay in order to attract applicants to the job. Thus, though difficult to assess, this situation can lead to a cost of not providing controls for occupational hazards.

Production Losses. Production losses can result from the absence of controls which cause occupational illness and injury. High turnover rate and employee absence can detrimentally affect production, resulting in a loss of profits to the organization. In addition, because of legal action or a feeling that the company is indifferent to the needs of the worker, employee morale and general attitude can become a problem which in turn may affect production.

Public Relations. The lack of necessary controls to protect the worker from occupational hazards can detrimentally affect the public image of the organization. This detrimental effect on the organization can result in a loss of sales, which affects the profitability of the firm.

Social Costs. Though not directly related to the profitability of the firm, there are certain social costs that result from a lack of control of occupational hazards. The first loss is to the family in terms of the loss of a wage earner. This results in a cost to the government to support families whose source of earning power has been removed. Though not directly a cost to an individual organization, industry in total helps to support such costs through taxes.

The Assessment of Risk

In order to determine the necessity for controls, the industrial hygiene engineer must assess the risks involved and the extent of illness or injury

which will result from an absence of control. This assessment is necessary because it is impossible to control all possible exposures to the worker. To do so would result in an astronomical cost. In general, the principle that should be used is to control the highest risk exposures first. The control of those exposures with the greatest loss potential will minimize the costs incurred as a result of not controlling the exposures.

For example, consider an organization employing 3500 employees with the following exposures:

Exposure A
Loss per occurrence is estimated at $1,000 with a probability of exposure of 1 in 100 for a given year.

Exposure B
Loss per occurrence is estimated at $10,500 with a probability of exposure of 1 in 500 for a year.

Exposure C
Loss per occurrence is estimated at $250,000 with a probability of exposure of 1 in 2500 for a year.

The question is, "Which exposure should be controlled first?" For the purpose of this illustration, it is assumed that the same cost is necessary to institute the controls for each of the three situations. In this example, the following calculations can be made:

Expected Loss (A) = $1,000 x .01 x 3500
Expected Loss (A) = $35,000 per year

Expected Loss (B) = $10,500 x .002 x 3500
Expected Loss (B) = $73,500 per year

Expected Loss (C) = $250,000 x .0004 x 3500
Expected Loss (C) = $350,000 per year

In the above example, it can be seen that the control of Exposure C should be undertaken first. If sufficient dollars are available to control another exposure, then Exposure B should be controlled. Finally, when dollars are available to control Exposure A, this can be done. Although the above analysis was based upon pure economics and did not consider the potential exposure level directly, an indirect consideration of the exposure level was made in terms of the cost per occurrence. It is likely that a higher cost per occurrence is a result of the more serious loss, thus indirectly the seriousness of the exposure is considered.

The Economic Cost of Providing Industrial Hygiene Controls

On the other side of the coin, certain costs are incurred as the result of providing industrial hygiene control. The following discussion centers around these costs.

Equipment Costs. The installation of industrial hygiene controls may require equipment. The cost of equipment includes not only the capital expenditure for the equipment itself but also any delivery and installation costs as well as the space that is used by the equipment. Purchasing and engineering costs must be considered. Any modifications that must be made to the plant and other production equipment to accommodate the control equipment must also be considered to be a cost of control.

Effects on Production. The installation of industrial hygiene controls can have a detrimental effect on the production of the plant. First, there is a potential for lost production during the installation of the controls themselves. In addition, any downtime necessary for the maintenance of the controls may result in the necessity to shut down production. Finally, in some cases, the controls themselves may limit the productive capacity of the facility. Such a situation might be the case where special handling of toxic materials is required and where such handling is accomplished at a rate slower than the process equipment using the materials.

Costs of Control Operation. Many of the controls which are used to protect workers from occupational hazards involve mechanical and electrical equipment. This equipment does not operate without a cost. The cost of the power to operate the equipment as well as the cost of maintenance in terms of both hours and replacement parts must be considered. Any personal protective equipment for maintenance and production personnel, which is necessary should the controls fail, must also be considered.

Cost of Training. When an industrial hygiene control has been installed, it is necessary to provide training to the workers in precautions concerning the use of the control equipment. Training in the operation of the equipment as well as training of the maintenance crew is also necessary. This training involves the man hours of both a trainer and the crews and, as a result, is a cost of installing the control.

Equipment Life. Once the control has been put in place, it in effect becomes a necessary part of the production facility. At some time, this equipment must be replaced. Replacement of this equipment is a cost of providing control in the first case.

Other Related Costs. Other costs that are related to the provision of industrial hygiene controls involve special packaging which may be required to protect workers from toxic substances, warning or monitoring equipment which may be necessary to indicate when the control is no longer working, and waste removal costs for toxic substances which have been removed from the work area.

Not all costs will be incurred for all methods of control. The costs for various controls will differ. For example, the operating cost may be high while the cost of obtaining and installing the equipment is low. Such would be the case in terms of a dilution ventilation system where the installation and purchase of such a system is relatively low while the operating cost is high. On the other hand, in the case of a local exhaust system, the opposite may be the case. The purchase and installation of a local exhaust system usually involves a high capital expenditure while the operating costs are low relative to a dilution ventilation system.

Selecting a Control Method

Various general methods are available for use in the control of occupational hazards. These methods include:

1. Elimination
2. Substitution
3. Process or procedure change
4. Isolation or enclosure of the source or worker
5. Ventilation
6. Administrative control
7. Personal protective equipment

Within each of these general methods of control, various alternative approaches are available for use to control a particular occupational hazard. It is important to consider the various alternatives in some logical manner to assure that the best choice is made for control of hazards. In order to do this, it is important that a logical and systematic approach be used. Such a systematic procedure involves a number of steps, which are discussed below.

Identify the Exposure. The first step in selecting a control method to be implemented for a particular occupational hazard is to identify the hazard exposure. To identify the hazard exposure, a number of questions should be answered. What is the exposure? Who is exposed, when are they exposed, where does the exposure occur, and how does the exposure occur? What is the extent of the hazard involved? Is the loss severe, moderate, or low? What is the estimated cost of the loss? What risks are involved with the exposure? Is the likelihood of exposure high or low? What personnel are subjected to the exposure when it occurs? What specifically is the objective of the control method?

By answering these questions, the industrial hygiene engineer has clearly defined the problem that is faced and identified the result which he wishes to obtain. The importance of specifying the result that is desired cannot be overemphasized. It is difficult to compare various methods of control when the desired result is unknown.

Choosing Alternatives. After the problem has been clearly identified and the objectives for control stated, the industrial hygiene engineer should identify those alternatives that can be implemented to obtain the stated objective. It cannot be emphasized too much that the first choice alternative should not be selected without comparing this choice to other alternatives. Often the first choice is not the best method for control, either in terms of the desired result or in terms of the cost involved. For this reason, it is necessary that alternative methods for control be considered.

As a starting point for identifying various alternatives, consider the seven general methods of control previously listed. Each of these general methods of control can act as an organizer from which alternative control procedures may be identified. Within each general method of control, specific alternatives can be identified.

Once the initial list of alternatives has been developed, the search process should not stop. It may be possible to improve upon the list of alternatives by identifying modifications to the selected alternatives. By modifying the alternatives, new and better alternatives may result.

When an alternative has been identified, the possibility of modifying it to obtain another alternative can be accomplished by the following:

1. Consider if any part of the alternatives can be eliminated.
2. Can two or more alternatives be combined?
3. Can the procedures for use be changed?
4. Can changes be made in some parts of the alternative to make it more attractive?

By answering these and other questions, it may be possible to improve upon the alternative list used for comparison.

One technique that can be utilized to generate alternatives involves a group process. Based upon the old adage that two heads are better than one and that there is a synergistic effect within a group, the technique of brainstorming may be used to identify alternative methods of control. Brainstorming operates on the principle that, initially, the quantity of ideas is more important than quality. Ideas are not evaluated until an exhaustive list has been developed. Then, by either combining or modifying the various approaches suggested during the brainstorming session, perhaps a new and more creative alternative, resulting in better control for a lower cost, can be developed.

Predicting Results. After the list of alternative controls has been developed, it will be obvious that not all alternatives are equal in terms of either their effects or their costs of implementation. To determine which alternative is best, it is important to be able to predict the results of selecting and implementing a particular alternative control. The prediction can be made as a result of experience. If experience with a particular control has not been obtained, the experience of others or experimental results may provide the data necessary for the prediction. The objective of such prediction is to determine as much as possible the result that will be obtained by instituting the particular alternative control; that is, how much will the risk factor be lowered? Also, what will be the cost of implementing the alternative?

Evaluation of Alternatives. After predictions have been made for the results of each of the alternatives, it is necessary to evaluate the alternatives to determine the best or most acceptable alternative. The most important question to ask is, "What alternatives will result in reaching the objective stated for the control of the hazard?" By asking this question, the list of alternatives can be pared down through the elimination of those alternatives that do not meet the desired criteria stated in the objective.

The next step is to determine, from those alternatives which attain the objective, the one that provides the best economic advantage. Some basic

techniques of economic analysis can be used to carry out this step. These techniques are discussed in the next chapter.

Finally, once the alternatives have been compared in terms of the desired result and the economics of implementation, the decision can be made concerning which alternative provides the best approach to controlling the occupational hazard. In most cases, the decision will be obvious, based on the use of the appropriate analysis techniques.

Summary

In this chapter, the cost of industrial hygiene control has been discussed from two points of view. From the first point of view, the cost of not providing controls for occupational hazards has been considered. Among the costs that are important are the increased costs of workers' compensation, production losses, training and recruitment costs, and medical and disability premium increases. When it is determined that an industrial control is desired, certain costs necessary to implement the control are incurred. Among the costs involved are the cost of the equipment, its operation and maintenance, as well as any production losses that may result from the installation of the equipment in the production process.

The industrial hygiene engineer is often tempted to recommend the installation of controls whenever a worker is exposed to an occupational hazard. Though such a recommendation certainly is to be applauded in terms of the desired outcome, that of a safe and healthful workplace, the facts do not always indicate that management will take the same viewpoint. The industrial hygiene engineer must be aware of the manager's view that to operate an industrial concern involves certain risks that must be taken. As a result, it becomes incumbent to sell the need for such controls to management. In order to perform this job in a credible fashion, the industrial hygiene engineer must approach the institution of a particular control from the viewpoint that management is likely to take. This includes the need to consider the economic costs involved in providing the control versus those involved with not providing the control. Though this situation is not ideal from the standpoint of the occupational safety and health professional, realization of the need to sell industrial hygiene control may, in the long run, maximize the desired objective--that of providing a safe and healthful environment for the worker.